

ECC

AIM

To implement **Elliptic Curve Cryptography (ECC)** using ECDH to generate a secure shared secret between two users.

ALGORITHM

1. Start the program.
2. Generate an ECC key pair for **Alice**.
3. Generate an ECC key pair for **Bob**.
4. Exchange the public keys.
5. Compute the shared secret using each user's private key and the other's public key.
6. Compare the generated shared secrets.
7. Display the shared secret and verify that both are equal.
8. Stop the program.

PROGRAM

```
import java.security.KeyPair;
import java.security.KeyPairGenerator;
import java.util.Arrays;
import javax.crypto.KeyAgreement;

public class Main {

    public static void main(String[] args) throws Exception {

        KeyPairGenerator keyGen = KeyPairGenerator.getInstance("EC");
        keyGen.initialize(256);

        KeyPair alicePair = keyGen.generateKeyPair();
        KeyPair bobPair = keyGen.generateKeyPair();

        KeyAgreement alice = KeyAgreement.getInstance("ECDH");
```

```

    alice.init(alicePair.getPrivate());

    alice.doPhase(bobPair.getPublic(), true);

    byte[] aliceSecret = alice.generateSecret();


    KeyAgreement bob = KeyAgreement.getInstance("ECDH");

    bob.init(bobPair.getPrivate());

    bob.doPhase(alicePair.getPublic(), true);

    byte[] bobSecret = bob.generateSecret();


    System.out.println("Alice Shared Secret:");

    for (byte b : aliceSecret)

        System.out.printf("%02X", b);


    System.out.println("\n\nBob Shared Secret:");

    for (byte b : bobSecret)

        System.out.printf("%02X", b);

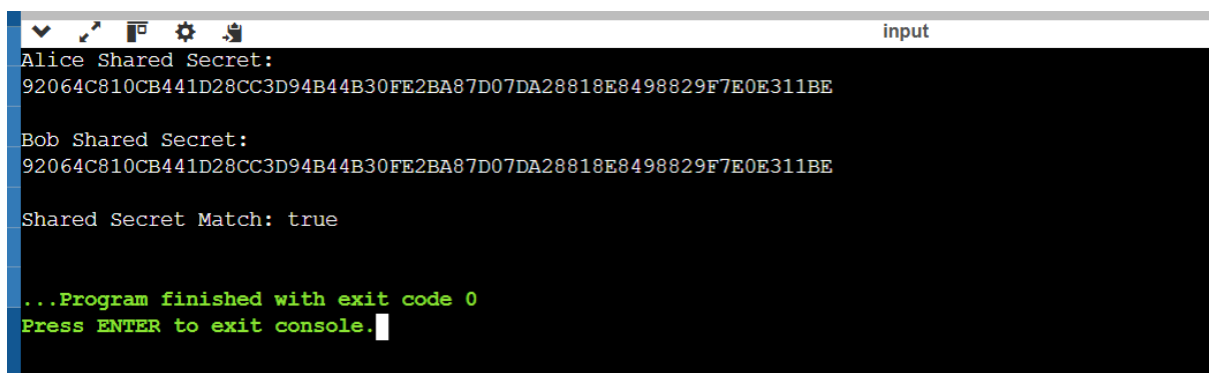

    System.out.println("\n\nShared Secret Match: " +

        Arrays.equals(aliceSecret, bobSecret));

}

}

```



```

input
Alice Shared Secret:
92064C810CB441D28CC3D94B44B30FE2BA87D07DA28818E8498829F7E0E311BE

Bob Shared Secret:
92064C810CB441D28CC3D94B44B30FE2BA87D07DA28818E8498829F7E0E311BE

Shared Secret Match: true

...Program finished with exit code 0
Press ENTER to exit console.

```

RESULT

Thus, **ECC using ECDH** was implemented successfully, and both users generated the same shared secret.

